Waveform Design for Watermark Framework Based DFRC System With Application on Joint SAR Imaging and Communication

Jing Yang, Youshan Tan, *Student Member, IEEE*, Xianxiang Yu, Guolong Cui, *Senior Member, IEEE*, and Di Zhang, *Senior Member, IEEE*

***Abstract*— In this article, the watermarking framework for electromagnetic systems is established with nonblind, semiblind, and blind watermarking demodulation processes mitigated to applications like radar detection, synchronization, integrated dual function, etc. Desired for similar advantages and trade-offs of watermarking technology, the dual function radar and communication (DFRC) system is specifically concerned for covert communication and low possibility of interception (LPI) radar sensing. In this respect, we propose a novel DFRC waveform design method where peak sidelobe level (PSL) of autocorrelation function (ACF) is considered as the figure of merit with information embedded via discrete Fourier transform (DFT) watermarking strategy. Meanwhile, the peak-to-average ratio (PAR) and energy constraints are forced to ensure compatibility with current hardware technique. To handle the resulting NP-hard design problems, the proximal method of multipliers (PMM) is employed with the overall computational burden linear with the amount of information per pulse and quadratic with respect to the code length. Finally, numerical and experimental results are provided to evaluate the effectiveness of the proposed DFRC waveform design scheme with application in joint synthetic aperture radar (SAR) imaging and communication.**

***Index Terms*— Dual function radar and communication (DFRC) system, peak sidelobe level (PSL), proximal method of multipliers (PMM), synthetic aperture radar (SAR) imaging, watermarking framework.**

NOMENCLATURE

Bold letters	$\boldsymbol{a}$ (lower case), and $\boldsymbol{A}$ (upper case) denote vector and matrix, respectively.
$(\cdot)^T$, $(\cdot)^*$, and $(\cdot)^\dagger$	Transpose, the conjugate, and the conjugate transpose operators, respectively.
$\mathbb{C}^{N\times M}$, $\mathbb{R}^N$, and $\mathbb{C}^N$	Sets of $N \times M$-dimensional matrices of complex and N-dimensional vectors of real and complex numbers, respectively.
$\|\boldsymbol{s}\|$	Euclidean norm.
$\boldsymbol{I}_N$	$N \times N$-dimensional identity matrix.
$\boldsymbol{0}_N \in \mathbb{R}^N$	Vector with all entries being 0.
$\boldsymbol{1}$	Vector with all entries being 1.
Letter j	Imaginary unit (i.e., $j = \sqrt{-1}$).
$\Re\{\cdot\}$ and $\Im\{\cdot\}$	Real and imaginary parts of a complex number, respectively.
$\arg(\cdot)$ and $\|\cdot\|$	Argument and modulus of a complex valued scalar, vector or matrix.
$E[\cdot]$ and $\odot$	Statistical expectation and the Hadamard product, respectively.
$\boldsymbol{a}/\boldsymbol{b}$	$\boldsymbol{a}/\boldsymbol{b} \triangleq [a(1)/b(1), \ldots, a(N)/b(N)]^T \in \mathbb{C}^N$.
$\text{sign}(\cdot)$	Signum function.

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Jing Yang and Di Zhang are with the School of Electrical and Information Engineering, Zhengzhou University, Zhengzhou 450001, China (e-mail: yangjinguestc@163.com; iedzhang@zzu.edu.cn).

Youshan Tan, Xianxiang Yu, and Guolong Cui are with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: 18782296203@163.com; xianxiangy@gmail.com; cuiguolong@uestc.edu.cn).

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I. INTRODUCTION

MOTIVATED by new opportunities for beyond 5G and 6G networks brought by the integrating hardware system and sharing signal processing modules, joint radar and communication (JRC), also known as integrated sensing and communications (ISAC), is emerging as a hot research topic [1], [2], [3], which includes the radar-communication coexistence (RCC) design to solve the spectrum congestion problem [4], [5], [6], [7] and dual function radar and communication (DFRC) scheme for cooperation and interaction [8]. Therein, DFRC waveform design has drawn flourishing attention. It plays a key role for mutual assistance of radar and communication functionalities, which is the main concern in this study.

One convenient approach is embedding communication symbols to the inherent parameters of existing signaling strategies, e.g., linear frequency modulation (LFM), frequency-modulated continuous waveform (FMCW), frequency hopping (FH), and orthogonal frequency division multiplexing (OFDM). Zhipeng et al. [9] and Nowak et al. [10] embed communication information in LFM signal via minimum shift keying (MSK) and M-sequences, respectively, which are capable of stable sensing with low data rate. In [11],

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index modulation by carrier and antenna selection, waveform permutation, and phase modulation (PM) is adopted for information transmission based on FMCW-MIMO radar scheme. Baxter et al. [12] introduce a generalized FH-based communication embedding model including FH permutation, frequency carrier/FH index modulation and PM, which incorporates most existing FH-MIMO DFRC systems [13], [14], [15]. In [16], the weighted sum of the communications data information rate (DIR) and the conditional mutual information (MI) based on the integrated OFDM waveform is maximized for the robust design without restriction on peak-to-average ratio (PAR).

Some researches also focus on designing electromagnetic distinctions of DFRC waveform in transformation domain for communication modulation while ensuring radar performance. Therein, Cui et al. [17] and Yang et al. [18] address the local ambiguity function (AF) optimization with information embedded via frequency and AF nulling, respectively. Invoking spatio-spectral modulation strategy to communicate with multiusers, Yu et al. [19] propose an optimization synthesis method of multiinput multioutput (MIMO) waveforms for a desired transmit beampattern with PAR, power, communication, and the main lobe width constraints restricted.

In addition, some studies synthesize the beamforming weight vectors of the individual orthogonal waveforms with an intrinsic geometric structure of communication channel for MIMO DFRC system, where amplitude shift keying (ASK) [20], phase shift keying (PSK) [21], quadrature amplitude modulation (QAM) [22], and permutation [23], [24] techniques are exploited for communication at the matched filter output of the users in the expected directions. On the contrary to [20], [21], [22], [23], and [24], Liu et al. [25] design the beamforming weight vectors of the data streams with radar estimation performance and communication quality of service (QoS) guaranteed by Cramér-Rao bound minimization and signal-to-interference-plus-noise ratio (SINR) restrictions for each communication user, respectively. Aiming at minimizing the multiuser interference for communication while formulating a beampattern similar to the reference code, Liu et al. [26] propose waveform design procedures with total and per-antenna power constraint or constant modulus constraint, while [27] on that basis further considers the range sidelobe control. However, PAR constraint is not considered in [20], [21], [22], [23], [24], [25], and [27], where power efficiency for detection cannot be assured for practical implementations.

In light of the state-of-the-art literature about DFRC waveform design, manifold dual function waveform construction methods in various system frameworks are elaborated with different objectives and constraints for the trade-off between radar/sensing and communication performance. The following question arises: is there a general framework that can accommodate most DFRC systems? Besides, a small amount of work is applied to synthetic aperture radar (SAR) imaging, with most focusing on detection or estimation. In [28], mutually orthogonal waveforms are simultaneously transmitted by different channels and beams for joint MIMO-SAR imaging and wireless communication, respectively. Utilizing direct sequence spread spectrum (DSSS) waveforms, Herschfelt and Bliss [29] develop multiple access communications with a joint receiver for SAR imaging simultaneously. Zhang and Xia [30] insert a sufficient cyclic prefix to each OFDM pulse for higher range resolution in the SAR system. However, Wang et al. [28], Herschfelt and Bliss [29], and Zhang and Xia [30] adopt the communication-centric design over an existing communication waveform without a specific consideration in the trade-off of SAR imaging and communication performance [3].

Aimed at filling these gaps, this article proposes a general watermarking framework for various radio applications with particular attention paid to the DFRC system. In a departure from most DFRC approaches, a novel DFRC waveform design method with information embedded via the discrete Fourier transform (DFT) watermarking strategy is proposed while suppressing the peak sidelobe level (PSL) of autocorrelation function (ACF). The contributions of this work are highlighted as follows.

1) *General Watermarking Framework With Different Demodulation Schemes:* Different electromagnetic functions [e.g., radar, channel estimation, synchronization, identification friend or foe (IFF), integrated dual function] are summarized into the framework of watermarking technology for the first time with nonblind, semiblind, or blind watermarking demodulation process according to whether the original data or the embedded watermark is required in the decoder. In particular, the watermark framework of a complete DFRC system is especially specified demanding similar advantages and trade-offs of watermark technology.
2) *DFRC Waveform Design Method via DFT Watermarking Strategy:* A novel DFRC waveform design scheme is proposed based on the DFT watermarking strategy while suppressing the PSL of ACF. Furthermore, PAR and energy constraints are considered to comply with the current hardware technology. To handle the resulting NP-hard design problems, the proximal method of multipliers (PMM) is introduced, where closed-form solutions can be derived in each iteration with the converged sequence proved to be a Karush–Kuhn–Tucker (KKT) point. Besides, its computational complexity is linear with the amount of information per pulse and quadratic with respect to the code length.
3) *Numerical and Experimental Simulations Validation:* To shed light on the capability of the devised DFRC waveform design scheme to realize a trade-off between radar and communication functions, numerical simulations and experimental investigations are carried out with some analysis on ACF, DFT value, computational complexity, bit error rate (BER), etc.

Results highlight the effectiveness of the proposed framework intuitively through its application on SAR imaging while transmitting an image to users.

This article is organized as follows. In Section II, a general watermark framework suitable for electromagnetic system models with different functions is addressed with specified demodulation processes. Section III focuses on the DFRC system and proposes the signal modeling and problem formation of the integrated waveform. In Section IV, an iterative

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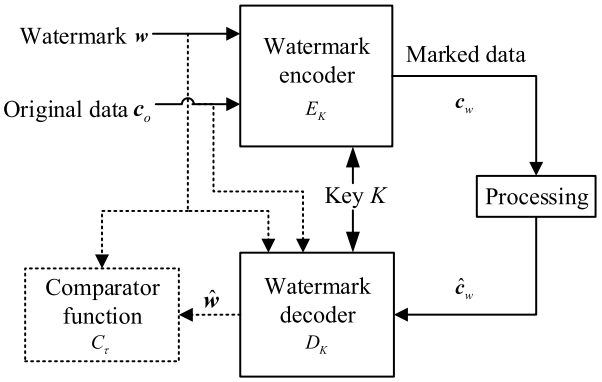


Fig. 1. General watermark framework.

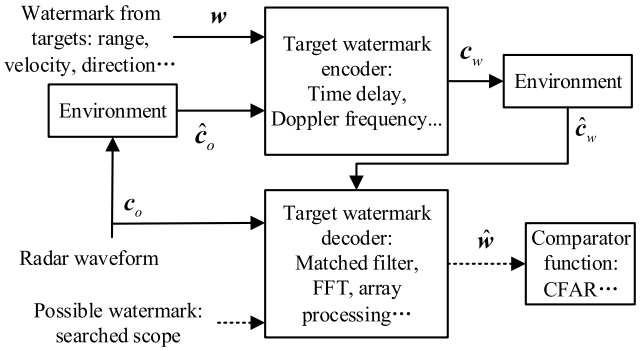


Fig. 2. Nonblind watermarking block diagram for radar model.

algorithm leveraging PMM is devised to tackle the resulting optimization problem. Section V presents some numerical and experimental results to assess the DFRC performance. Finally, in Section VI, concluding remarks and some possible future research avenues are provided.

II. System Model Based on Watermark Framework

In this section, let us focus on the electromagnetic system model based on the watermark framework with different functions, which is probably capable of radar detection, synchronization, IFF, DFRC, and so on.

A. General Digital Watermarking Technology

A general watermark model whose framework shown in Fig. 1 can be described by a tuple [31]

$$\mathcal{S} = \langle \mathcal{W}, \mathcal{O}, \mathcal{K}, E_K, D_K, C_\tau \rangle$$

where

1) $\mathcal{W}$ set of all watermarks with the payload message to be embedded;
2) $\mathcal{O}$ set of all carrier (original) data;
3) $\mathcal{K}$ set of all keys, which is capable of encryption to prevent the watermark being intercepted;
4) E_K embedding process, which can be represented as a function

$$E_K(\boldsymbol{c}_o, \boldsymbol{w}) = \boldsymbol{c}_w \tag{1}$$

with a secret or public key $K \in \mathcal{K}$, where $\boldsymbol{w} \in \mathcal{W}$ is the watermark to be embedded, $\boldsymbol{c}_o \in \mathcal{O}$ and $\boldsymbol{c}_w$ are the carrier data and marked data, respectively;

5) D_K demodulation process, which can be further categorized according to whether the original data $\boldsymbol{c}_o$ or watermark $\boldsymbol{w}$ is required;
6) C_τ comparator function, which contrasts the extracted watermark $\widehat{\boldsymbol{w}}$ with the suspected one against a threshold τ, i.e.,

$$C_\tau(\widehat{\boldsymbol{w}}, \boldsymbol{w}) = \begin{cases} 1, & c \geq \tau \\ 0, & c < \tau. \end{cases}$$

Digital watermarking technology is widely used to hide information in multimedia data for copyright protection or authentication. It is worth noting that watermarking technology has been applied to the radar and sonar waveform preliminarily to obtain the advantages like robustness, transparency, and security. Therein, Zhang et al. [32] and Bao et al. [33] propose the radar watermark waveform design methods where extra phases are embedded as watermarks to discriminate the jamming and transmitted signal. In [34] and [35], authentication codes are embedded in the original LFM sonar signal by the discrete cosine transform (DCT) watermarking method for origin identification. Remarkably, these embedded imperceptible signatures enable decreasing the probability of interception, IFF, and resistance to deceptive interference by introducing anticounterfeit and encryption techniques of watermarking in radar or sonar sensing. However, the related works only invoke some simple methods from watermarking technology for identification and confronting deception jamming. To the best of the authors' knowledge, there is no relevant research on applying watermarking technology to sensing waveforms for ISAC or other functions.

In Sections II-B–II-E, different radio function models (e.g., radar detection, synchronization, and dual function system) are migrated to a digital watermarking framework with specified demodulation processes, where the similarities of different systems and advantages brought by the corresponding watermarking architecture are discussed.

B. Nonblind Watermarking

Original data is required for the nonblind watermarking technique in the detection process which can be expressed as

$$D_K(\hat{\boldsymbol{c}}_w, \boldsymbol{c}_o) = \widehat{\boldsymbol{w}} \tag{2}$$

or

$$D_K(\hat{\boldsymbol{c}}_w, \boldsymbol{c}_o, \boldsymbol{w}) = \widehat{\boldsymbol{w}} \tag{3}$$

with a potential comparator function

$$C_\tau(\widehat{\boldsymbol{w}}) = \begin{cases} 1, & c \geq \tau \\ 0, & c < \tau \end{cases} \tag{4}$$

where $\hat{\boldsymbol{c}}_w$ and $\widehat{\boldsymbol{w}}$ represent the received marked data and the demodulated watermark, respectively.

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Application Scenario Migration: The radar detection process can be regarded as a nonblind watermarking technique, whose framework is illustrated in Fig. 2. Different from the active modulation of digital watermarking, the target information is passively embedded in the radar waveform (original data) via watermark in the form of time delay, Doppler frequency, etc. Then, the detection method to estimate corresponding parameters can be regarded as the decoder, where the watermark may be required for searching within a certain scope (e.g., direction). Without loss of generality, let us take the range measurement as an example. The range of a target (watermark $\boldsymbol{w}$) is encoded into the transmit waveform (original data $\boldsymbol{c}_o$) via time delay (encoder) and decoded via the matched filter (decoder with known $\boldsymbol{c}_o$). Then, with the aid of a threshold derived by the constant false alarm rate detector (comparator function C_τ), the range cell can be extracted.

Another typical example of nonblind watermarking is channel estimation realized by transmitting pilot signals known to both the user and base station. Then, the channel state information can be estimated by measuring the received signal within a pilot resource [36]. Furthermore, some communication-centered DFRC systems that directly use communication signals for sensing [30], [37], [38] can be treated as the estimation of the channel between integrated platform-target-integrated platform for sensing while communicating with users. Hence, their signal processing for target sensing is also a nonblind watermarking procedure.

C. Semiblind Watermarking

The decoding process of semiblind watermarking requires the embedded watermark instead of original data, which can be expressed as

$$D_K(\hat{\boldsymbol{c}}_{\boldsymbol{w}}, \boldsymbol{w}) = \widehat{\boldsymbol{w}} \tag{5}$$

and a comparator function may be further adopted to judge whether the demodulated watermark $\widehat{\boldsymbol{w}}$ belongs to the set of permitted watermarks $\mathcal{W}$, i.e.,

$$C_\tau(\widehat{\boldsymbol{w}}, \boldsymbol{w}) = \begin{cases} 1, & c \geq \tau \\ 0, & c < \tau. \end{cases} \tag{6}$$

Application Scenario Migration: Semiblind watermarking is widely used in copy control and copyright protection, which can be extended to the framework of IFF, synchronization, requesting detection, and so on. In [39], a watermark is embedded into the short-time Fourier transform (STFT) of sonar waveform for identification and then demodulated by a correlation detector which is a semiblind decoding process. Besides, Fig. 3(a) and (b) are the typical block diagrams of the carrier and timing synchronization, respectively [40]. Specifically, carrier synchronization is widely required in radio systems, where the carrier frequency can be viewed as a watermark that is embedded by the frequency upconversion of a mixer, and then demodulated by downconversion to obtain the original baseband waveform for subsequent signal processing. As to timing synchronization for the single carrier, a fixed synchronization code is hidden in the transmit waveform to mark the start transiting time. Then, by correlating the known

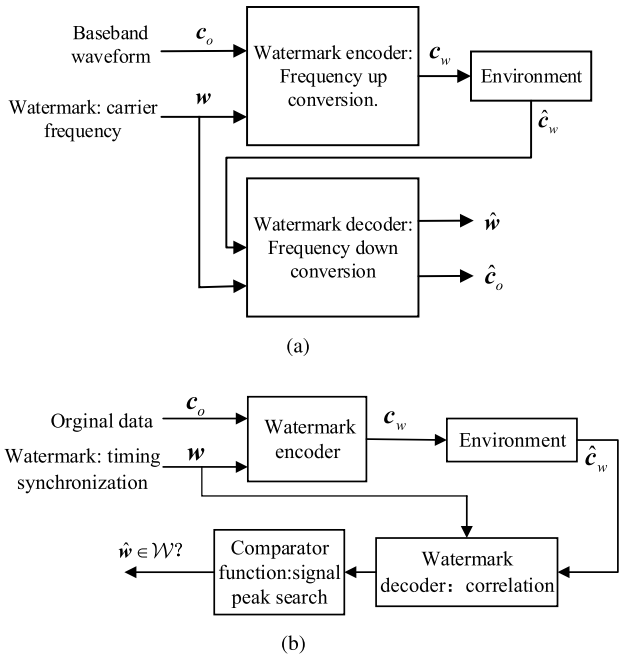


Fig. 3. Semiblind watermarking frameworks for (a) carrier synchronization and (b) timing synchronization.

synchronization sequence with the received signal samples at different sampling times, the arisen signal energy peak is searched by the comparator function, which can serve as the initial instant estimate.

D. Blind Watermarking

In the blind watermarking procedure, both the original data and the watermarking information of the demodulation process are unknown, which can be declared as

$$D_K(\hat{\boldsymbol{c}}_w) = \widehat{\boldsymbol{w}} \tag{7}$$

with a comparator function potentially further implemented, i.e.,

$$C_\tau(\widehat{\boldsymbol{w}}) = \begin{cases} 1, & c \geq \tau \\ 0, & c < \tau. \end{cases} \tag{8}$$

Application Scenario Migration: The integrated dual function system can be viewed as communication of watermark for the secondary function, while consisting of the original data for the primary function (see [17], [18], [19], [20], [21], [22], [23], [24]). Taking the DFRC and dual function jamming and communication (DFJC) systems as typical examples, the communication information is encoded into the joint waveform via time, frequency, beampattern, etc. In this respect, the radar or jamming waveform represents the original data with communication being the watermark. The baseband transmit waveform is totally unknown to the communication user equipments (UEs), which conforms to the characteristics of the blind watermark framework. With the decoder (and the keys), the UEs can demodulate and recover the transmitted messages.

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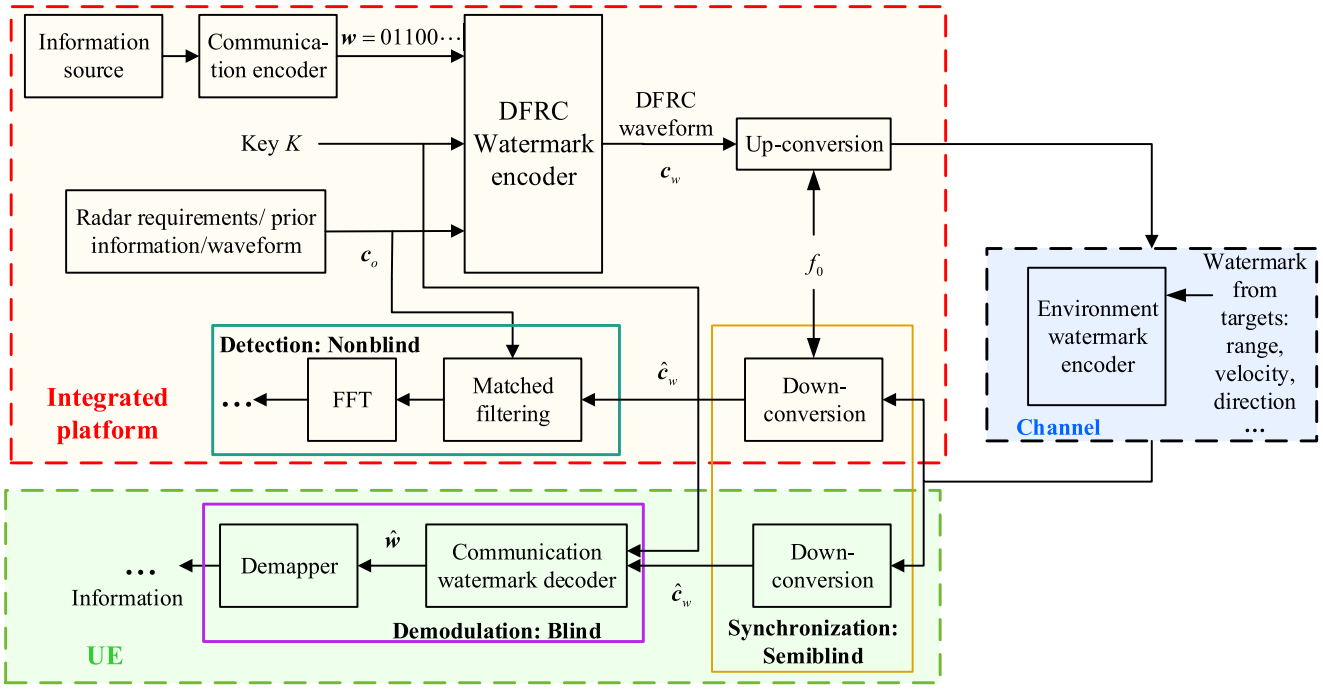


Fig. 4. Block diagram for DFRC system.

E. Watermarking Framework Involved in a DFRC System

Let us consider a DFRC system that consists of an integrated platform for joint radar sensing and information transmission, and the UEs for information reception. As shown in Fig. 4, the DFRC system mainly involves the following:

1) the design of dual function waveform, up and down-conversion, signal processing for radar sensing at the integrated platform;
2) downconversion, communication demodulation at the UEs' receiver end.

Based on the discussions above, the dual function waveform design, upconversion, and the embedding of target parameters can be regarded as the watermark encoder, while the signal processing for detection, downconversion, and communication can be concerned as nonblind, semiblind, and blind watermark decoder, respectively.

Aiming at secrecy communications and sensing simultaneously, the DFRC system is expected to have similar advantages of watermarking technology as follows.

1) *Robustness*: The resistance ability for geometric, temporal, volumetric distortions (e.g., scaling, delay, quantization), and attacks (e.g., jamming) is required to prevent the changes and modifications of the embedded information (e.g., target parameter and communication as the watermark) during the transmission of the integrated waveform and demodulation process.
2) *Perceptual Transparency*: The encoding algorithm must ensure the quality of the underlying host data with the embedded watermark which is nonperceptible or invisible for the signal processing of the primary function. Taking [23] as an example, the secondary communication function is transparent for radar beampattern performance in the radar-centric DFRC system.
3) *Steganography*: Confidential communication required by military and civilian applications can be realized through tailored embedding mechanisms [18], [19] and cryptographic means.
4) *Reversible Self-Recovery* [41]: Although the integrated waveform is not precisely known at the UEs' receiver end, the communication information should be extracted by the demodulation algorithm while maintaining integrity and validity.

Trade-offs for the DFRC systems similar to the watermarking scheme should also be attained between three mutually exclusive factors [41].

1) *Robustness.*
2) *Quality*: The demodulation quality of embedded information (for sensing or communication) is mainly determined by the distortion of the marked data and energy of watermark.
3) *Capacity*: Number of symbols or bits transmitted in the carrier data within fixed time

which is a challenging issue. In this regard, some methods in watermarking technology are of referential significance for the DFRC waveform design. Unlike general watermarking techniques, the original waveform of a dual function system can also be tailored for low possibility of interception (LPI) radar sensing [42], [43], which inspire us to design $\boldsymbol{c}_w$ directly. In Section III, a novel DFRC signal design method is proposed with communication information embedded in DFT domain as the watermark.

III. Signal Modeling and Problem Formulation

In this section, a DFRC signal model is developed with communication information embedded by DFT watermarking strategy. The optimization problem is formulated taking into

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account minimizing PSL of the ACF for radar detection with communication, energy, and PAR constraints.

A. Signal Modeling

Let us consider the dual-function signal $\boldsymbol{s} \in \mathbb{C}^N$. The ACF of $\boldsymbol{s}$ can be defined as

$$r_k = \sum_{n=1}^{M-k} s_n s_{n+k}^*. \tag{9}$$

Denoting the ACF vector by $\boldsymbol{r} = [r_0, r_1, \ldots, r_{N-1}, \boldsymbol{0}_{M-2N+1}^T, r_{1-N}, \ldots, r_{-1}]^T \in \mathbb{C}^M$ with $M \geq 2N-1$ representing its length, $\boldsymbol{r}$ can be efficiently computed by fast Fourier transform (FFT) and inverse FFT (IFFT)

$$\boldsymbol{r} = \boldsymbol{F}\left\{\left(\boldsymbol{F}^\dagger \bar{\boldsymbol{s}}\right) \odot \left(\boldsymbol{F}^\dagger \bar{\boldsymbol{s}}\right)^*\right\} \tag{10}$$

where $\bar{\boldsymbol{s}} = [\boldsymbol{s}; \boldsymbol{0}_{M-N}]$, $\boldsymbol{F} = [\boldsymbol{f}_0, \boldsymbol{f}_1, \ldots, \boldsymbol{f}_{M-1}] \in \mathbb{C}^{M\times M}$ is the DFT matrix with $\boldsymbol{f}_m = (1/(M)^{1/2})[1, e^{j2\pi(m-1/M)}, \ldots, e^{j2\pi(M-1)(m-1/M)}]^T$.

B. Communication Rule

Let us embed communication information by the DFT value of $\boldsymbol{s}$ [44] which can be expressed as

$$\boldsymbol{g} = \boldsymbol{F}_2^\dagger \boldsymbol{s} \in \mathbb{C}^{M_2} \tag{11}$$

with the length $M_2 \geq N$, $\boldsymbol{F}_2 = [\bar{\boldsymbol{f}}_0, \bar{\boldsymbol{f}}_1, \ldots, \bar{\boldsymbol{f}}_{M_2-1}] \in \mathbb{C}^{N\times M_2}$, and $\bar{\boldsymbol{f}}_m = (1/(M_2)^{1/2})[1, e^{j2\pi(m-1/M_2)}, \ldots, e^{j2\pi(N-1)(m-1/M_2)}]^T$.

Assume the embedded information $w_i \in \{1, -1\}$ is embedded in the l_ith spectral point with the following rule.

1) If $w_i = 1$, the imaginary part of $g(l_i)$ is 0, i.e., $g(l_i) = g(l_i)^*$.
2) If $w_i = -1$, the real part of $g(l_i)$ is 0, i.e., $g(l_i) = -g(l_i)^*$

which can be rewritten as

$$\boldsymbol{g}_c - \boldsymbol{w} \odot \boldsymbol{g}_c^* = \boldsymbol{0}_L \tag{12}$$

where $\boldsymbol{g}_c = \boldsymbol{F}_c^\dagger \boldsymbol{s} = [g(l_1), \ldots, g(l_L)]^T \in \mathbb{C}^L$ with $\boldsymbol{F}_c = [\bar{\boldsymbol{f}}_{l_1}, \ldots, \bar{\boldsymbol{f}}_{l_L}] \in \mathbb{C}^{N\times L}$, and $\boldsymbol{w} = [w_1, \ldots, w_L]^T \in \mathcal{W}$ is the communication watermark. Assigning $\boldsymbol{l} = [l_1, \ldots, l_L]$ the vector of positions with $0 \leq l_1 < \cdots < l_L \leq M_2 - 1$ and $L \leq M_2$, $\boldsymbol{l}$ is benefit to achieve steganography for the secret communication, which can be viewed as the key in the watermarking technology, i.e., $\boldsymbol{l} \in \mathcal{K}$. As a result, assuming the pulse repetition frequency (PRF) of DFRC system is f_{PRF}, it will deliver L bits information in each pulse with the communication rate of

$$R_b = L \times f_{\text{PRF}}. \tag{13}$$

The received signal at the UE's receiver end can be written as

$$\boldsymbol{x} = \alpha_0 \boldsymbol{s} + \boldsymbol{n} \tag{14}$$

where α_0 is the complex channel coefficient representing the propagation environment from the integrated platform to the UE, and $\boldsymbol{n}$ denotes the Gaussian additive noise with zero mean and covariance $\sigma^2 \boldsymbol{I}_N$. In this respect, the signal-to-noise ratio (SNR) is defined as $10\log_{10}(E[\|\alpha_0 \boldsymbol{s}\|^2]/N\sigma^2)$ dB. It follows that the DFT value of $\boldsymbol{x}$ at the preassigned grids can be computed by:

$$\bar{\boldsymbol{g}}_c = \boldsymbol{F}_c^\dagger \boldsymbol{x} = \alpha_0 \boldsymbol{F}_c^\dagger \boldsymbol{s} + \boldsymbol{F}_c^\dagger \boldsymbol{n}. \tag{15}$$

Assuming that α_0 is estimated accurately, the communication information can be demodulated by comparing the amplitude of the real and imaginary parts of $\tilde{\boldsymbol{g}}_c = \bar{\boldsymbol{g}}_c/\alpha_0$.

1) If $|\Re\{\tilde{g}_c(i)\}| \geq |\Im\{\tilde{g}_c(i)\}|$, the demodulated information is set $\tilde{w}_i = 1$.
2) If $|\Re\{\tilde{g}_c(i)\}| < |\Im\{\tilde{g}_c(i)\}|$, the demodulated information is set $\tilde{w}_i = -1$.

C. Problem Formulation

To assure compatibility with the amplifier technology and radar power budget, energy and PAR constraints are forced on the sought waveform, respectively,

$$\|\boldsymbol{s}\|^2 = N \tag{16}$$

$$\frac{\max_{n=1,\ldots,N} |s(n)|^2}{\frac{1}{N}\|\boldsymbol{s}\|^2} \leq \gamma. \tag{17}$$

Substituting into the energy constraint (16), (17) can be further transformed to $|s(n)| \leq (\gamma)^{1/2}$. In particular, PAR constraint reduces to the constant modulus constraint setting $\gamma = 1$.

Besides, the amplitude of $\boldsymbol{g}_c$ is limited to $|g_c(i)| \in [u_1, u_2], i = 1, \ldots, L$ to ensure the communication demodulation performance with u_1, u_2 the lower and upper bounds, respectively.

For both radar target detection and SAR imaging, it is desirable to design waveforms with low ACF sidelobes to ensure correlation performance of the receiver [45], [46]. To this end, the PSL of ACF of the transmitted waveform $\boldsymbol{s}$ is minimized with communication information embedded. Hence, the optimization problem can be posed as

$$\mathcal{P}_0 \begin{cases} \min\limits_{\boldsymbol{s}} & \max\limits_n |r_n|_{n\in\Theta_m} \\ \text{s.t.} & \|\boldsymbol{s}\|^2 = N \\ & |s(n)| \leq \sqrt{\gamma}, \quad n = 1, \ldots, N \\ & \boldsymbol{g}_c - \boldsymbol{w} \odot \boldsymbol{g}_c^* = \boldsymbol{0}_L, \\ & |g_c(i)| \in [u_1, u_2], \quad i = 1, \ldots, L \end{cases} \tag{18}$$

where Θ_m denotes the autocorrelation regions to be minimized.

IV. Dual-Function Signal Synthesis

Introducing auxiliary variables $\boldsymbol{y} = \boldsymbol{F}^\dagger \bar{\boldsymbol{s}}, \boldsymbol{r} = \boldsymbol{F}\{|\boldsymbol{F}^\dagger \bar{\boldsymbol{s}}| \odot |\boldsymbol{y}|\}$ and $\boldsymbol{g}_c = \boldsymbol{F}_c^\dagger \boldsymbol{s}$ with $\bar{\boldsymbol{s}} = [\boldsymbol{s}; \boldsymbol{0}_{M-N}]$, $\mathcal{P}_0$ can be equivalently transformed to

$$\mathcal{P}_1 \begin{cases} \min\limits_{\boldsymbol{s},\boldsymbol{y},\boldsymbol{r}} & \max\limits_n |r_n|_{n\in\Theta_m} \\ \text{s.t.} & \|\boldsymbol{s}\|^2 = N \\ & |s(n)| \leq \sqrt{\gamma}, \quad n = 1, \ldots, N \\ & \boldsymbol{g}_c - \boldsymbol{w} \odot \boldsymbol{g}_c^* = \boldsymbol{0}_L \\ & |g_c(i)| \in [u_1, u_2], \quad i = 1, \ldots, L \\ & \boldsymbol{y} = \boldsymbol{F}^\dagger \bar{\boldsymbol{s}} \\ & \boldsymbol{r} = \boldsymbol{F}\left\{|\boldsymbol{F}^\dagger \bar{\boldsymbol{s}}| \odot |\boldsymbol{y}|\right\} \\ & \boldsymbol{g}_c = \boldsymbol{F}_c^\dagger \boldsymbol{s}. \end{cases} \tag{19}$$

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The proximal augmented Lagrangian of $\mathcal{P}_1$ can be constructed as

$$\begin{aligned}
&L(\boldsymbol{s},\boldsymbol{y},\boldsymbol{r},\boldsymbol{g}_c,\boldsymbol{\lambda},\boldsymbol{\mu},\boldsymbol{\nu})\\
&=\max_n\{|r_n|\}_{n\in\Theta_m}\\
&\quad+\frac{1}{2\gamma_r}\left|\left|\boldsymbol{r}-\boldsymbol{r}^{(t)}\right|\right|_2^2+\frac{1}{2\gamma_y}\left|\left|\boldsymbol{y}-\boldsymbol{y}^{(t)}\right|\right|_2^2+\frac{1}{2\gamma_g}\left|\left|\boldsymbol{g}_c-\boldsymbol{g}_c^{(t)}\right|\right|_2^2\\
&\quad+\frac{\rho_1}{2}\left(\left|\left|\boldsymbol{r}-\boldsymbol{F}\left(|\boldsymbol{F}^\dagger\bar{\boldsymbol{s}}|\odot|\boldsymbol{y}|\right)+\boldsymbol{\lambda}\right|\right|_2^2-\left|\left|\boldsymbol{\lambda}\right|\right|_2^2\right)\\
&\quad+\frac{\rho_2}{2}\left(\left|\left|\boldsymbol{y}-\boldsymbol{F}^\dagger\bar{\boldsymbol{s}}+\boldsymbol{\mu}\right|\right|_2^2-\|\boldsymbol{\mu}\|_2^2\right)\\
&\quad+\frac{\rho_3}{2}\left(\left|\left|\boldsymbol{g}_c-\boldsymbol{F}_c^\dagger\boldsymbol{s}+\boldsymbol{\nu}\right|\right|_2^2-\|\boldsymbol{\nu}\|_2^2\right)
\end{aligned}$$

where

1) γ_r, γ_y, and γ_g proximal parameters;
2) ρ_1, ρ_2, and ρ_3 penalty parameters;
3) $\boldsymbol{\lambda}$, $\boldsymbol{\mu}$, and $\boldsymbol{\nu}$ scaled dual variables;
4) $\boldsymbol{r}^{(t)}$, $\boldsymbol{y}^{(t)}$, and $\boldsymbol{g}_c^{(t)}$ tth iteration of $\boldsymbol{r}$, $\boldsymbol{y}$, and $\boldsymbol{g}_c$, respectively.

Next, PMM procedure [47] is employed and reported in Algorithm 1, where $(\boldsymbol{s},\boldsymbol{y},\boldsymbol{r},\boldsymbol{g}_c,\boldsymbol{\lambda},\boldsymbol{\mu},\boldsymbol{\nu})$ is determined in an iterative fashion. The analytic proof of the converged sequence being a KKT point under mild conditions is provided in [47]. The following subsections pave the way to the solutions to problems (20)–(23).

Algorithm 1 PMM Procedure for $\mathcal{P}_1$

Require: $\boldsymbol{y}^{(0)},\boldsymbol{r}^{(0)},\boldsymbol{g}_c^{(0)},\boldsymbol{\lambda}^{(0)},\boldsymbol{\mu}^{(0)},\boldsymbol{\nu}^{(0)},\gamma_r,\gamma_y,\gamma_g,\rho_1,\rho_2,\rho_3,\boldsymbol{F}_c$;
Ensure: DFRC waveform $\boldsymbol{s}^{(t+1)}$;

1: $t=0$;
2: Update $\boldsymbol{s}^{(t+1)},\boldsymbol{y}^{(t+1)},\boldsymbol{r}^{(t+1)},\boldsymbol{g}_c^{(t+1)}$ by solving the following problems:

$$\begin{aligned}
\boldsymbol{s}^{(t+1)}:=&\arg\min_{\boldsymbol{s}} L(\boldsymbol{s},\boldsymbol{y}^{(t)},\boldsymbol{r}^{(t)},\boldsymbol{g}_c^{(t)},\boldsymbol{\lambda}^{(t)},\boldsymbol{\mu}^{(t)},\boldsymbol{\nu}^{(t)})\\
&\text{s.t. }\|\boldsymbol{s}\|^2=N,\\
&\quad|s(n)|\le\sqrt{\gamma},n=1,\cdots,N; \qquad (20)\\
\boldsymbol{y}^{(t+1)}:=&\arg\min_{\boldsymbol{y}} L(\boldsymbol{s}^{(t+1)},\boldsymbol{y},\boldsymbol{r}^{(t)},\boldsymbol{g}_c^{(t)},\boldsymbol{\lambda}^{(t)},\boldsymbol{\mu}^{(t)},\boldsymbol{\nu}^{(t)}); \qquad (21)\\
\boldsymbol{r}^{(t+1)}:=&\arg\min_{\boldsymbol{r}} L(\boldsymbol{s}^{(t+1)},\boldsymbol{y}^{(t+1)},\boldsymbol{r},\boldsymbol{g}_c^{(t)},\boldsymbol{\lambda}^{(t)},\boldsymbol{\mu}^{(t)},\boldsymbol{\nu}^{(t)}); \qquad (22)\\
\boldsymbol{g}_c^{(t+1)}:=&\arg\min_{\boldsymbol{g}_c} L(\boldsymbol{s}^{(t+1)},\boldsymbol{y}^{(t+1)},\boldsymbol{r}^{(t+1)},\boldsymbol{g}_c,\boldsymbol{\lambda}^{(t)},\boldsymbol{\mu}^{(t)},\boldsymbol{\nu}^{(t)})\\
&\text{s.t. }\boldsymbol{g}_c-\boldsymbol{w}\odot\boldsymbol{g}_c^*=\boldsymbol{0}_L,\\
&\quad|g_c(i)|\in[u_1,u_2],i=1,\ldots,L; \qquad (23)
\end{aligned}$$

3: Update $\boldsymbol{\lambda}^{(t+1)}$, $\boldsymbol{\mu}^{(t+1)}$ and $\boldsymbol{\nu}^{(t+1)}$ by:

$$\begin{aligned}
\boldsymbol{\lambda}^{(t+1)}&:=\boldsymbol{\lambda}^{(t)}+\boldsymbol{r}^{(t+1)}-\boldsymbol{F}\{|\boldsymbol{F}^\dagger\bar{\boldsymbol{s}}^{(t+1)}|\odot|\boldsymbol{y}^{(t+1)}|\}; \qquad (24)\\
\boldsymbol{\mu}^{(t+1)}&:=\boldsymbol{\mu}^{(t)}+\boldsymbol{y}^{(t+1)}-\boldsymbol{F}^\dagger\bar{\boldsymbol{s}}^{(t+1)}; \qquad (25)\\
\boldsymbol{\nu}^{(t+1)}&:=\boldsymbol{\nu}^{(t)}+\boldsymbol{g}_c^{(t+1)}-\boldsymbol{F}_c^\dagger\boldsymbol{s}^{(t+1)}; \qquad (26)
\end{aligned}$$

4: If the exit condition is satisfied, output $\boldsymbol{s}^{(t+1)}$. Otherwise, $t:=t+1$, and return to Step 2;

A. Update of $\boldsymbol{s}^{(t+1)}$

Ignoring the irrelevant constant terms with $\boldsymbol{F}\boldsymbol{F}^\dagger=\boldsymbol{I}_M$, problem (20) can be equivalently transformed to

$$\begin{aligned}
\min_{\boldsymbol{s}}\quad&\chi(\bar{\boldsymbol{s}})\\
\text{s.t.}\quad&\bar{\boldsymbol{s}}=\left[\boldsymbol{s};\boldsymbol{0}_{\mathrm{M-N}}\right]\\
&\|\boldsymbol{s}\|^2=N\\
&|s(n)|\le\sqrt{\gamma},\quad n=1,\ldots,N \qquad (27)
\end{aligned}$$

where

$$\chi(\bar{\boldsymbol{s}})=\frac{\rho_1}{2}\chi_1(\bar{\boldsymbol{s}})+\frac{\rho_2}{2}\chi_2(\bar{\boldsymbol{s}})+\frac{\rho_3}{2}\chi_3(\bar{\boldsymbol{s}}) \qquad (28)$$

$$\chi_1(\bar{\boldsymbol{s}})=\left|\left||\boldsymbol{F}^\dagger\bar{\boldsymbol{s}}|\odot|\boldsymbol{y}^{(t)}|-\Re\left\{\boldsymbol{F}^\dagger\bar{\boldsymbol{\lambda}}^{(t)}\right\}\right|\right|_2^2 \qquad (29)$$

$$\chi_2(\bar{\boldsymbol{s}})=\left|\left|\boldsymbol{F}^\dagger\bar{\boldsymbol{s}}-\overline{\boldsymbol{\mu}}^{(t)}\right|\right|_2^2 \qquad (30)$$

$$\chi_3(\bar{\boldsymbol{s}})=\left|\left|\boldsymbol{F}_c^\dagger\boldsymbol{s}-\overline{\boldsymbol{\nu}}^{(t)}\right|\right|_2^2 \qquad (31)$$

with $\bar{\boldsymbol{\lambda}}^{(t)}=\boldsymbol{r}^{(t)}+\boldsymbol{\lambda}^{(t)}$, $\overline{\boldsymbol{\mu}}^{(t)}=\boldsymbol{y}^{(t)}+\boldsymbol{\mu}^{(t)}$, and $\overline{\boldsymbol{\nu}}^{(t)}=\boldsymbol{g}_c^{(t)}+\boldsymbol{\nu}^{(t)}$.

Let us first find the majorization function of $\chi_1(\bar{\boldsymbol{s}})$. Resorting to Cauchy–Schwartz equality and $\Re\{\boldsymbol{y}^\dagger\boldsymbol{s}\}\le|\boldsymbol{y}^\dagger\boldsymbol{s}|$, it can be derived that

$$\begin{aligned}
\chi_1(\bar{\boldsymbol{s}})&\le\sum_{m=1}^{M}\left||y_m^{(t)}|\boldsymbol{f}_m^\dagger\bar{\boldsymbol{s}}-z_m^{(t)}\right|^2\\
&=\left|\left|\left(\boldsymbol{F}^\dagger\bar{\boldsymbol{s}}\right)\odot|\boldsymbol{y}^{(t)}|-\boldsymbol{z}^{(t)}\right|\right|_2^2\\
&=\overline{\chi}_1(\bar{\boldsymbol{s}}) \qquad (32)
\end{aligned}$$

with $\boldsymbol{z}^{(t)}=\Re\{\boldsymbol{F}^\dagger\bar{\boldsymbol{\lambda}}^{(t)}\}\odot e^{j\arg(\boldsymbol{F}^\dagger\bar{\boldsymbol{s}}^{(t)})}$.

Dividing $\boldsymbol{F}=[\boldsymbol{F}_1;\boldsymbol{F}_2]$ with $\boldsymbol{F}_1\in\mathbb{C}^{N\times M}$ and $\boldsymbol{F}_2\in\mathbb{C}^{(M-N)\times M}$, it follows that:

$$\begin{aligned}
\frac{\rho_1}{2}\overline{\chi}_1(\bar{\boldsymbol{s}})+\frac{\rho_2}{2}\chi_2(\bar{\boldsymbol{s}})&=\left|\left|\boldsymbol{a}_1^{(t)}\odot\left(\boldsymbol{F}^\dagger\bar{\boldsymbol{s}}-\boldsymbol{a}_2^{(t)}\right)\right|\right|_2^2+\hat{c}\\
&=\left|\left|\boldsymbol{a}_1^{(t)}\odot\left(\boldsymbol{F}_1^\dagger\boldsymbol{s}-\boldsymbol{a}_2^{(t)}\right)\right|\right|_2^2+\hat{c}\\
&\le\left|\left|\boldsymbol{a}_1^{(t)}\right|\right|_2^2\left|\left|\boldsymbol{F}_1^\dagger\boldsymbol{s}-\boldsymbol{a}_2^{(t)}\right|\right|_2^2+\hat{c} \qquad (33)
\end{aligned}$$

where $\boldsymbol{a}_1^{(t)}=((\rho_1/2)|\boldsymbol{y}^{(t)}|^2+(\rho_2/2))^{1/2}$, $\boldsymbol{a}_2^{(t)}=(\rho_1|\boldsymbol{y}^{(t)}|\odot\boldsymbol{z}^{(t)}+\rho_2\overline{\boldsymbol{\mu}}^{(t)}/\rho_1|\boldsymbol{y}^{(t)}|^2+\rho_2)$, and $\hat{c}$ is a constant term irrelevant with $\boldsymbol{s}$.

Hence, (27) can be transformed to

$$\begin{aligned}
\min_{\boldsymbol{s}}\quad&\boldsymbol{s}^\dagger\boldsymbol{P}\boldsymbol{s}-\Re\left\{\boldsymbol{v}^{(t)\dagger}\boldsymbol{s}\right\}\\
\text{s.t.}\quad&\|\boldsymbol{s}\|^2=\mathrm{N}\\
&|s(n)|\le\sqrt{\gamma},\quad n=1,\ldots,N \qquad (34)
\end{aligned}$$

where $\boldsymbol{P}=\boldsymbol{F}_1\boldsymbol{F}_1^\dagger\|\boldsymbol{a}_1^{(t)}\|_2^2+(\rho_3/2)\boldsymbol{F}_c\boldsymbol{F}_c^\dagger$, and $\boldsymbol{v}^{(t)}=2\|\boldsymbol{a}_1^{(t)}\|_2^2\boldsymbol{F}_1\boldsymbol{a}_2^{(t)}+\rho_3\boldsymbol{F}_c\overline{\boldsymbol{v}}^{(t)}$.

According to majorization–minimization (MM) algorithm, a majorizer of the objective function of problem (34) can be constructed as follows [48, Lemma 1]:

$$\begin{aligned}
\hat{u}_1(\boldsymbol{s},\boldsymbol{s}^{(t)})=&\lambda_u(\boldsymbol{P})\boldsymbol{s}^\dagger\boldsymbol{s}+2\Re\left\{\boldsymbol{s}^\dagger(\boldsymbol{P}-\lambda_u(\boldsymbol{P})\boldsymbol{I})\boldsymbol{s}^{(t)}\right\}\\
&+\boldsymbol{s}^{(t)\dagger}(\lambda_u(\boldsymbol{P})\boldsymbol{I}-\boldsymbol{P})\boldsymbol{s}^{(t)}-\Re\left\{\boldsymbol{v}^{(t)\dagger}\boldsymbol{s}\right\}
\end{aligned}$$

with $\lambda_u(\boldsymbol{P})$ being the maximum eigenvalue of $\boldsymbol{P}$. As $\boldsymbol{P}$ is the sum of two Hermitian matrices, $\lambda_u(\boldsymbol{P})$ satisfies [49]

$$\begin{aligned}
\lambda_u(\boldsymbol{P})&\le\left|\left|\boldsymbol{a}_1^{(t)}\right|\right|_2^2\lambda_u\left(\boldsymbol{F}_1\boldsymbol{F}_1^\dagger\right)+\frac{\rho_3}{2}\lambda_u\left(\boldsymbol{F}_c\boldsymbol{F}_c^\dagger\right)\\
&=\left|\left|\boldsymbol{a}_1^{(t)}\right|\right|_2^2+\frac{\rho_3}{2}. \qquad (35)
\end{aligned}$$

To save the computational time, $\lambda_u(\boldsymbol{P})$ is simply chosen as $\|\boldsymbol{a}_1^{(t)}\|_2^2+(\rho_3/2)$.

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Since $\|\boldsymbol{s}\|^2 = N$, the majorized problem can be derived as

$$\begin{aligned} \max_{\boldsymbol{s}} \quad & \Re\{\bar{\boldsymbol{v}}^{(t)\dagger}\boldsymbol{s}\} \\ \text{s.t.} \quad & \|\boldsymbol{s}\|^2 = \mathrm{N} \\ & |s(n)| \le \sqrt{\gamma}, \quad n = 1, \ldots, N \end{aligned} \tag{36}$$

with $\bar{\boldsymbol{v}}^{(t)} = 2(\lambda_u(\boldsymbol{P})\boldsymbol{I} - \boldsymbol{P})\boldsymbol{s}^{(t)} + \boldsymbol{v}^{(t)}$, whose closed form solution is [50]

$$\boldsymbol{s}^{(t+1)} = \mathcal{P}_s\left(\bar{\boldsymbol{v}}^{(t)}\right) \tag{37}$$

where $|\bar{v}^{(t)}(1)| \ge |\bar{v}^{(t)}(2)| \ge \cdots \ge |\bar{v}^{(t)}(N)|$ is assumed without loss of generality with m the number of nonzero elements

$$\begin{aligned} \mathcal{P}_s(\cdot) = & \left(\mathbf{1}_{\mathbb{R}\ge 0}(N - m\gamma)\right)\sqrt{\gamma}\boldsymbol{u}_m \odot e^{j\arg(\cdot)} \\ & + \left(\mathbf{1}_{\mathbb{R}<0}(N - m\gamma)\right)\min\{\delta|\cdot|, \sqrt{\gamma}\mathbf{1}\} \odot e^{j\arg(\cdot)} \end{aligned} \tag{38}$$

$$\mathbf{1}_A(x) = \begin{cases} 1, & \text{if } x \in A \\ 0, & \text{else} \end{cases} \tag{39}$$

$$\boldsymbol{u}_m = \left[\underbrace{1, \ldots 1}_{m}, \underbrace{\sqrt{\frac{N - m\gamma}{(N-m)\gamma}}, \cdots \sqrt{\frac{N - m\gamma}{(N-m)\gamma}}}_{N-m}\right]^T$$

and

$$\begin{aligned} \delta \in \Bigg\{ \delta \Bigg| & \sum_{n=1}^{N} \min\left\{\delta^2\left|\bar{v}^{(t)}(n)\right|^2, \gamma\right\} = 1 \\ & \times \delta \in \left[0, \frac{\sqrt{\gamma}}{\min\left\{|\bar{v}^{(t)}(n)|, |\bar{v}^{(t)}(n)| \ne 0\right\}}\right] \Bigg\}. \end{aligned}$$

B. Update of $\boldsymbol{y}^{(t+1)}$

Problem (21) can be recast as

$$\min_{\boldsymbol{y}} g(\boldsymbol{y}) \tag{40}$$

where

$$\begin{aligned} g(\boldsymbol{y}) = & \frac{1}{\gamma_y}\left\|\boldsymbol{y} - \boldsymbol{y}^{(t)}\right\|_2^2 + \rho_2\left\|\boldsymbol{y} - \boldsymbol{F}^\dagger\overline{\boldsymbol{s}}^{(t+1)} + \boldsymbol{\mu}^{(t)}\right\|_2^2 \\ & + \rho_1\left\|\boldsymbol{F}\left(\left|\boldsymbol{F}^\dagger\overline{\boldsymbol{s}}^{(t+1)}\right| \odot |\boldsymbol{y}|\right) - \bar{\boldsymbol{\lambda}}^{(t)}\right\|_2^2. \end{aligned}$$

Using the same transformation strategy as (32) and (33)

$$\left\|\boldsymbol{F}\left(\left|\boldsymbol{F}^\dagger\overline{\boldsymbol{s}}^{(t+1)}\right| \odot |\boldsymbol{y}|\right) - \bar{\boldsymbol{\lambda}}^{(t)}\right\|_2^2 \le \left\|\left|\boldsymbol{F}^\dagger\overline{\boldsymbol{s}}^{(t+1)}\right| \odot \boldsymbol{y} - \boldsymbol{q}^{(t)}\right\|_2^2$$

where $\boldsymbol{q}^{(t)} = \Re\{\boldsymbol{F}^\dagger\bar{\boldsymbol{\lambda}}^{(t)}\} \odot e^{j\arg(\boldsymbol{y}^{(t)})}$.

In this respect, problem (40) can be transformed to

$$\min_{\boldsymbol{y}} \|\boldsymbol{y} - \tilde{\boldsymbol{y}}^{(t)}\|^2 \tag{41}$$

with $\tilde{\boldsymbol{y}}^{(t)} = (\rho_1|\boldsymbol{F}^\dagger\overline{\boldsymbol{s}}^{(t+1)}| \odot \boldsymbol{q}^{(t)} + (1/\gamma_y)\boldsymbol{y}^{(t)} + \rho_2(\boldsymbol{F}^\dagger\overline{\boldsymbol{s}}^{(t+1)} - \boldsymbol{\mu}^{(t)})/\rho_1|\boldsymbol{F}^\dagger\overline{\boldsymbol{s}}^{(t+1)}|^2 + (1/\gamma_y) + \rho_2)$, whose closed-form solution is

$$\boldsymbol{y}^{(t+1)} = \tilde{\boldsymbol{y}}^{(t)}. \tag{42}$$

C. Update of $\boldsymbol{r}^{(t+1)}$

Problem (22) can be recast as

$$\min_{\boldsymbol{r}} \max_{n}\{|r_n|\}_{n\in\Theta_m} + \frac{\zeta}{2}\left\|\boldsymbol{r} - \bar{\boldsymbol{b}}^{(t)}\right\|^2 \tag{43}$$

where $\zeta = (1/\gamma_r) + \rho_1$, $\boldsymbol{b}^{(t)} = \boldsymbol{F}(|\boldsymbol{F}^\dagger\overline{\boldsymbol{s}}^{(t+1)}| \odot \boldsymbol{y}^{(t+1)}) - \boldsymbol{\lambda}^{(t)}$, $\bar{\boldsymbol{b}}^{(t)} = (\frac{1}{\gamma_r}\boldsymbol{r}^{(t)} + \rho_1\boldsymbol{b}^{(t)}/\zeta)$.

Introducing an auxiliary variable η, problem (43) can be equivalently rewritten as

$$\begin{aligned} \min_{\boldsymbol{r},\eta} \quad & \eta + \frac{\zeta}{2}\left\|\boldsymbol{r} - \bar{\boldsymbol{b}}^{(t)}\right\|^2 \\ \text{s.t.} \quad & |r_n| \le \eta \quad \forall n \in \Theta_m. \end{aligned} \tag{44}$$

It is readily to derive that

$$\arg\left(\boldsymbol{r}^{(t+1)}\right) = \arg\left(\bar{\boldsymbol{b}}^{(t)}\right), \tag{45}$$

$$|r_n^{(t+1)}| = |\bar{b}_n^{(t)}|, \quad \text{if } n \notin \Theta_m. \tag{46}$$

Substituting (45) and (46) back into (44) yields

$$\begin{aligned} \min_{\eta,h_n} \quad & \eta + \frac{\zeta}{2}\sum_{n\in\Theta_m}\left(h_n - |\bar{b}_n^{(t)}|\right)^2 \\ \text{s.t.} \quad & 0 \le h_n \le \eta, \quad n \in \Theta_m \end{aligned} \tag{47}$$

with $h_n = |r_n|$, whose closed-form solution can be solved by the KKT theory given in [47]. Once $\eta^\star$ is given, h_n can be derived by

$$h_n^\star = \begin{cases} \eta^\star, & \text{if } \eta^\star \le |\bar{b}_n^{(t)}| \\ |\bar{b}_n^{(t)}|, & \text{otherwise.} \end{cases} \tag{48}$$

In this respect, we can drive that

$$|r_n^{(t+1)}| = h_n^\star, \quad \text{if } n \in \Theta_m. \tag{49}$$

D. Update of $\boldsymbol{g}_c^{(t+1)}$

Problem (23) with respect to $\boldsymbol{g}_c$ can be recast as

$$\begin{aligned} \min_{\boldsymbol{g}_c} \quad & \frac{\bar{\zeta}}{2}\left\|\boldsymbol{g}_c - \bar{\boldsymbol{d}}^{(t)}\right\|^2 \\ \text{s.t.} \quad & \boldsymbol{g}_c - \boldsymbol{w} \odot \boldsymbol{g}_c^* = \boldsymbol{0}_L \\ & |g_c(i)| \in [u_1, u_2], \quad i = 1, \ldots, L \end{aligned} \tag{50}$$

with $\bar{\zeta} = (1/\gamma_g) + \rho_3$, $\boldsymbol{d}^{(t)} = \boldsymbol{F}_c^\dagger\boldsymbol{s}^{(t)} - \boldsymbol{v}^{(t)}$, and $\bar{\boldsymbol{d}}^{(t)} = ((1/\gamma_g)\boldsymbol{g}^{(t)} + \rho_3\boldsymbol{d}^{(t)}/\bar{\zeta})$, whose closed- form solution is

$$g_c(m) = \begin{cases} \mathcal{P}_g\left(\Re\left\{\hat{d}^{(t)}(m)\right\}\right), & \text{if } w_i = \text{“1”} \\ j\mathcal{P}_g\left(\Im\left\{\hat{d}^{(t)}(m)\right\}\right), & \text{else} \end{cases} \tag{51}$$

with

$$\mathcal{P}_g(\cdot) = \mathrm{sign}(\cdot)\max(u_1, \min(|\cdot|, u_2)). \tag{52}$$

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E. Computational Complexity

Let us now analyze the computational complexity of Algorithm 1. $\boldsymbol{P}$ can be computed and saved before PMM algorithm starts with the complexity of $O((M+L)N^2)$. In each iteration, the update of $\boldsymbol{s}^{(t+1)}$ requires the computation of $\bar{\boldsymbol{v}}^{(t)}$ and (37). FFT and matrix-to-vector multiplications are implemented for $\bar{\boldsymbol{v}}^{(t)}$ with the total computational complexity of $O((M+L)N+N\log N)$. When $\gamma > 1$, δ is searched by the bisection method. For the convenience of analysis, we only focus on the constant envelope case for $\gamma = 1$. In this respect, the computational complexity of (37) is $O(N)$. The update of $\boldsymbol{y}^{(t+1)}$ relies on the computation of $\tilde{\boldsymbol{y}}^{(t)}$ with the complexity of $O(N\log N+N)$. The computational complexities of $\boldsymbol{r}^{(t+1)}$ and $\boldsymbol{g}_c^{(t+1)}$ are $O(N\log N+N)$ [47] and $O(LN)$, respectively. As a consequence, the total computational complexity of Algorithm 1 is $O((M+L)N^2+T_0N(M+L+\log N))$ with T_0 the number of iterations.

V. Performance Analysis

This section is devoted to the performance assessment of the proposed algorithm in terms of the ACF, BER, and computational complexity. Let us assign $N = 256$, $M = 2N$, $M_2 = N$, $\Theta_m = \{1, 2, \ldots, N-1\}$. For Algorithm 1, the exit condition $\epsilon^{(t)} = \|\boldsymbol{y}^{(t)} - \boldsymbol{F}^{\dagger}\bar{\boldsymbol{s}}^{(t)}\|^2 + \|\boldsymbol{r}^{(t)} - \boldsymbol{F}\{|\boldsymbol{F}^{\dagger}\bar{\boldsymbol{s}}^{(t)}| \odot |\boldsymbol{y}^{(t)}|\}\|^2 + \|\boldsymbol{g}_c^{(t)} - \boldsymbol{F}_c^{\dagger}\boldsymbol{s}^{(t)}\|^2 \le 10^{-6}$ and iteration number $\bar{T} \in [1000, 30000]$ are applied with penalty factor $\rho_1 = 1$, $\rho_2 = 10^3$, $\rho_3 = 10^8$, $\gamma_r = 10$, $\gamma_y = 10$ and $\gamma_g = 10^{-4}$. Unless otherwise stated, the dual variables and auxiliary variables are initialized by zero vectors. $\boldsymbol{l}$ is randomly picked among $\{1, \ldots, M_2\}$ and precisely known by the UE for communication demodulation.

A. Numerical and Experimental Assessment of Algorithm Performance

This section aims at showing a typical example of DFRC waveform synthesis and processing for the numerical and experimental performance assessment of the proposed PMM algorithm. To this end, "It is UESTC!" is set as the transmitted sentence for the current pulse, whose ASCII code is transformed to the binary code and then employed by $\boldsymbol{w}$. Hence, there are 12 symbols and $L = 84$ bits information in this pulse. Besides, $\boldsymbol{g}_c$ is required to satisfy the amplitude constraints with $u_1 = 0.5$ and $u_2 = 1.5$. Corresponding DFRC waveforms with $\gamma = 1, 2, 4$ are synthesized via Algorithm 1. To evaluate the engineering realizability of the proposed framework, the baseband waveforms are first upconverted, then generated by an arbitrary waveform generator with the carrier frequency $f_0 = 1.2$ GHz and sampling frequency $f_s = 3.2$ GHz, respectively. Transmitted with a microwave cable to the receive channel, the received signals are downconverted and processed. The actual scene and block diagram of semiphysical experiment platform are shown in Fig. 5(a) and (b), respectively. Therein, the ACF of the synthesized waveforms $\boldsymbol{s}$ derived by numerical simulations and cross correlation function (CCF) between $\boldsymbol{s}$ and recovered waveform received by the semiphysical experiment platform (i.e., the pulse compression result) are studied in the following.

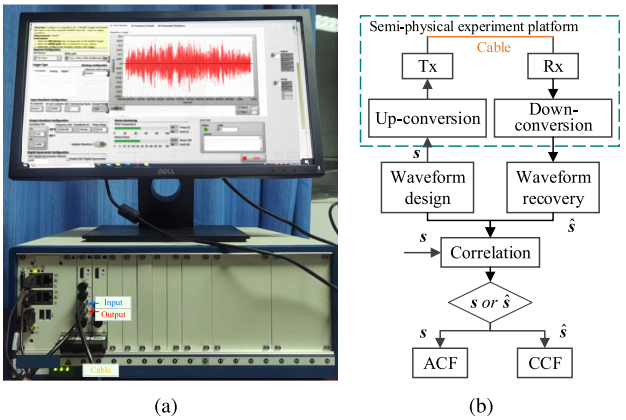


Fig. 5. Semiphysical experiment platform. (a) Photograph of the experimental scene. (b) Block diagram with emphasis on ACF and CCF.

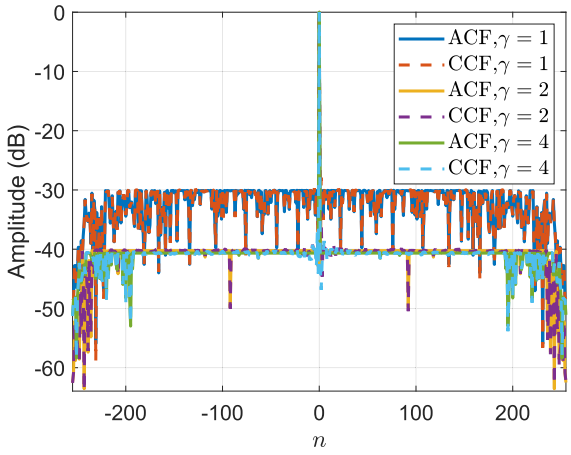


Fig. 6. Normalized ACF and CCF results derived by numerical simulations and experiments, respectively.

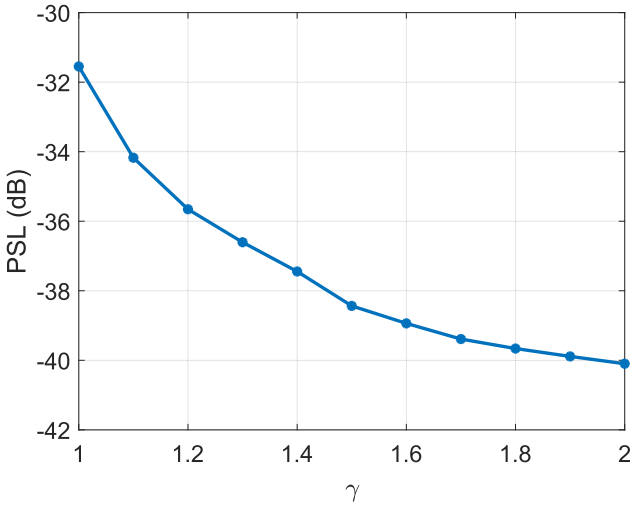


Fig. 7. PSL of ACF versus PAR value γ.

Fig. 6 depicts the normalized ACFs and CCFs of different PAR values with their PSL listed in Table I. As expected, increasing of γ provides a lower PSL being available more degrees of freedom (DoFs) at the design stage, whereas the improvement becomes subtle since $\gamma > 2$. The CCFs are

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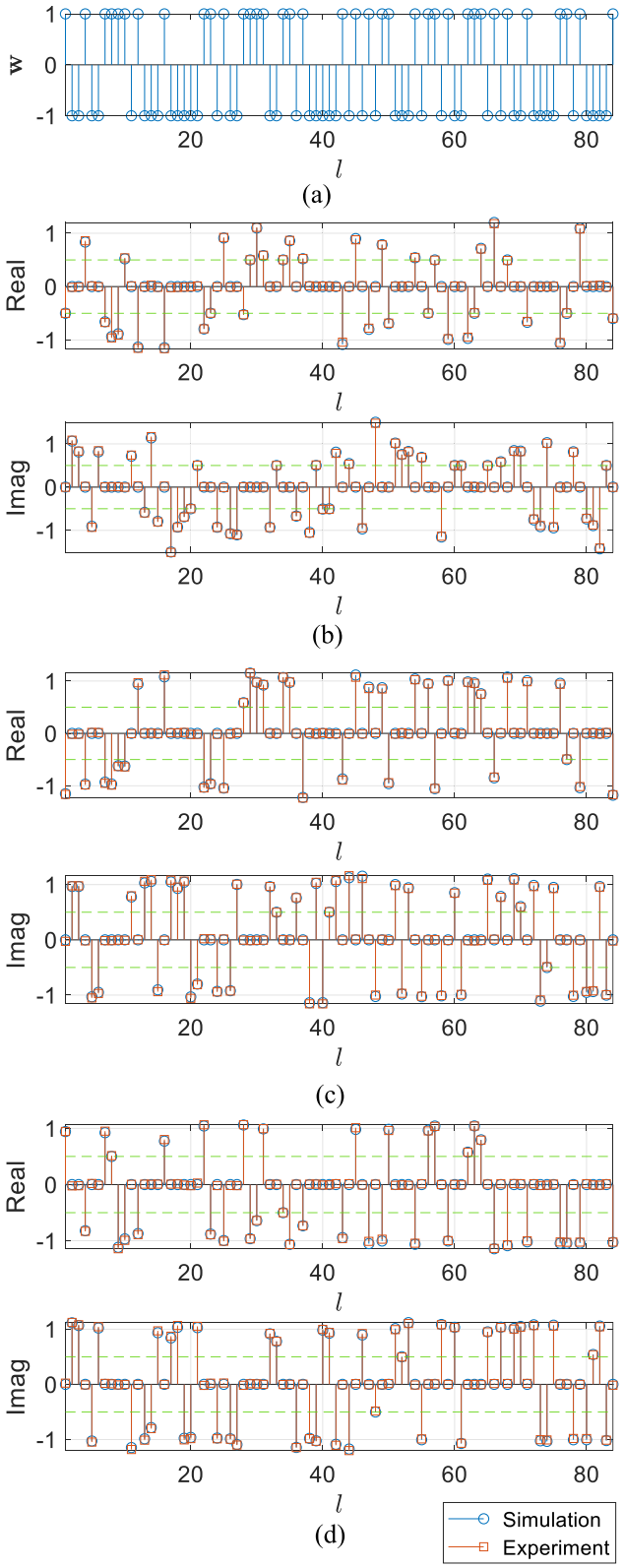


Fig. 8. Stem and leaf display of (a) $\boldsymbol{w}$ (desired information sequence) and real and imaginary parts of $\boldsymbol{g}_c$ and $\tilde{\boldsymbol{g}}_c$ for (b) $\gamma = 1$, (c) $\gamma = 2$, and (d) $\gamma = 4$.

close to their ACFs with low sidelobe levels generally.[1] Due to the limitations of the hardware platform as well as the digital carrier synchronization processing, the normalized PSL of experimental CCF lifts up for all γ compared with the simulation results. Furthermore, Fig. 7 shows the normalized PSL of ACF versus PAR value γ, which verifies again that the higher the γ, the lower PSL value.[2]

In Fig. 8(a)–(d), the stem and leaf display of $\boldsymbol{w}$, the real and imaginary parts of $\boldsymbol{g}_c$ derived by simulation, and $\tilde{\boldsymbol{g}}_c$ derived by experiments are provided with $\gamma = 1, 2, 4$, respectively. Therein, the green dashed lines in Fig. 8(b)–(d) denote the lower bound u_1 for $|g_c(i)|$. It is readily to find

[1]The SNR of the received signal without any processing is about 40 dB in these experiments.

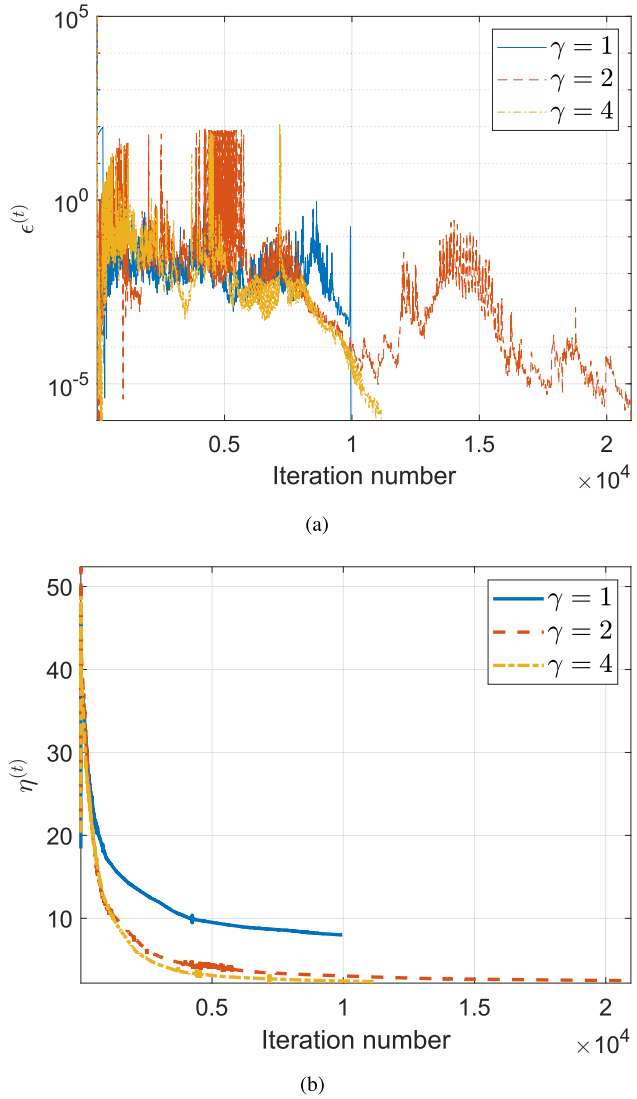


Fig. 9. (a) $\epsilon^{(t)}$ and (b) $\eta^{(t)}$ versus iteration number with different γ.

TABLE I
NORMALIZED PSL OF ACF AND CCF WITH DIFFERENT γ

PSL (dB)	$\gamma = 1$	$\gamma = 2$	$\gamma = 4$
ACF	-30.04	-40.24	-40.66
CCF	-27.93	-33.66	-37.07

[2]To ensure the PSL monotonically decreases versus PAR value, the optimized code at the previous PAR value (a grid of γ values, i.e., $1 = \gamma_1 \leq \gamma_2 \leq \cdots \leq \gamma_D = 2$, is supposed) is adopted for initialization.

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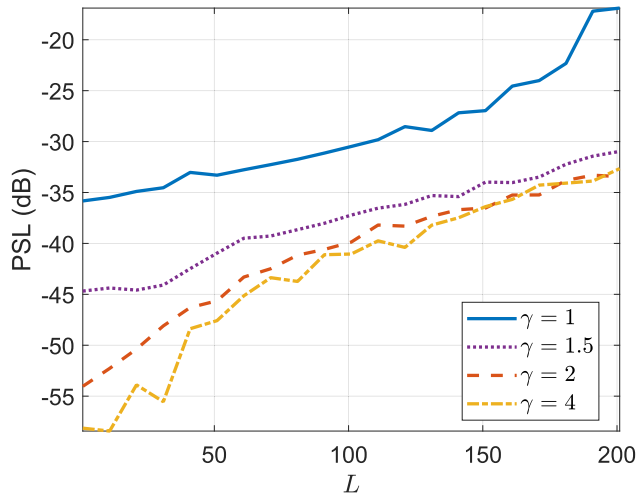


Fig. 10. Trade-off between radar and communication function with $u_1 = 0.5$ and $u_2 = 1.5$.

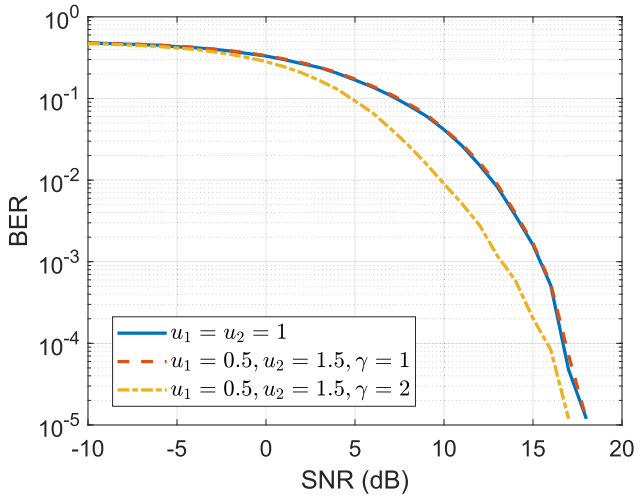


Fig. 11. BER versus SNR for different u_1, u_2, and γ.

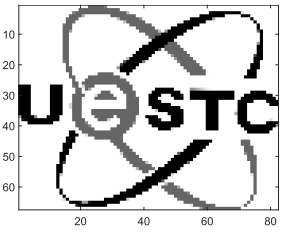


Fig. 12. Transmitted image for communication.

that the synthesized $\boldsymbol{g}_c$ by Algorithm 1 satisfies $\Re\{g_c(i)\} = 0$, $|\Im\{g_c(i)\}| \in [u_1, u_2]$ for $w_i =$"1," and $\Im\{g_c(i)\} = 0$, $|\Re\{g_c(i)\}| \in [u_1, u_2]$ for $w_i = -$"1" to modulate the communication information. $\tilde{\boldsymbol{g}}_c$ recovered by experiments are very close to $\boldsymbol{g}_c$ for different γ but with slightly deviation due to the device internal noise, quantization errors, and so on. As a result, the information code $\boldsymbol{w}$ is recovered without error in all experiments.

To shed light on the computational complexity and convergence performance, Fig. 9(a) and (b) represents feasibility

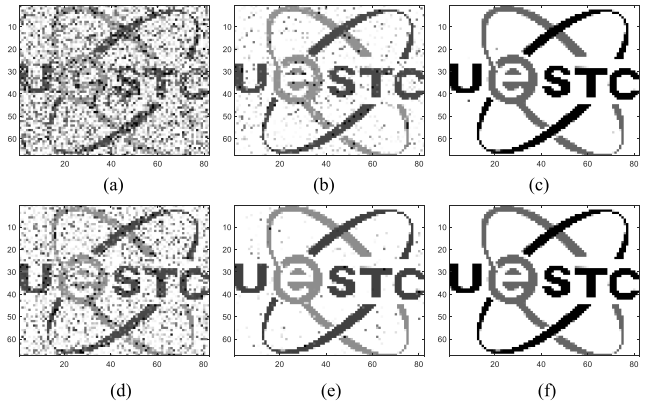


Fig. 13. Demodulated image for different SNR and γ. (a) SNR = 5 dB, $\gamma = 1$. (b) SNR = 11 dB, $\gamma = 1$. (c) SNR = 17 dB, $\gamma = 1$. (d) SNR = 5 dB, $\gamma = 2$. (e) SNR = 11 dB, $\gamma = 2$. (f) SNR = 17 dB, $\gamma = 2$.

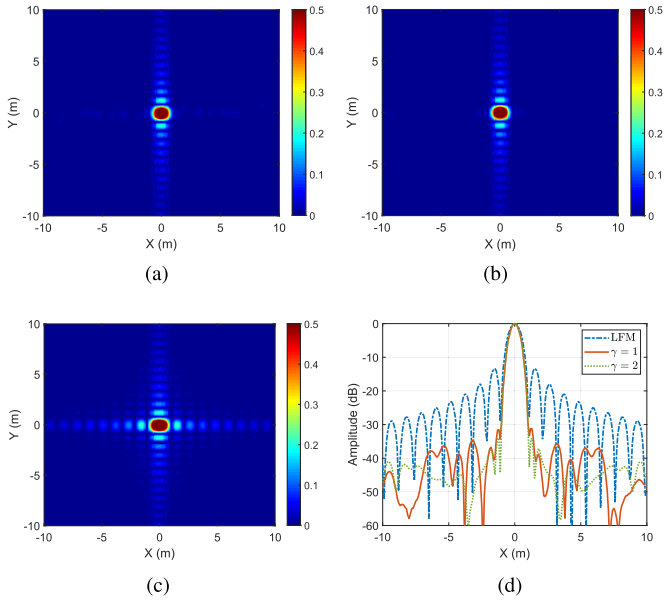


Fig. 14. Point target imaging performance of different signals and their range profiles along $Y = 0$ m. (a) DFRC signal with $\gamma = 1$. (b) DFRC signal with $\gamma = 2$. (c) LFM. (d) Range profile.

tolerances of primal residuals $\epsilon^{(t)}$ and PSL $\eta^{(t)}$ versus iteration number with different γ, respectively. Inspection of the curves shows that Algorithm 1 is capable of converging with PSL gradually suppressed, which provides a practical proof of the PMM framework efficiency.

B. Trade-off Between Radar and Communication Functions

To assess the trade-off of radar and communication performance, Fig. 10 depicts the lowest normalized PSL value among four Monte Carlo experiments versus the length of random information code $\boldsymbol{w}$ with $u_1 = 0.5$, $u_2 = 1.5$ for $\gamma = 1, 1.5, 2, 4$, respectively. Inspection of the plot clearly reveals that the PSLs of ACF generally rise with the increase of communication rate for different γ, namely the improvement of communication rate degrades the sensing performance. As expected, the larger the PAR parameter γ, the lower PSL for more DoFs at the design stage regardless of L. Besides, the

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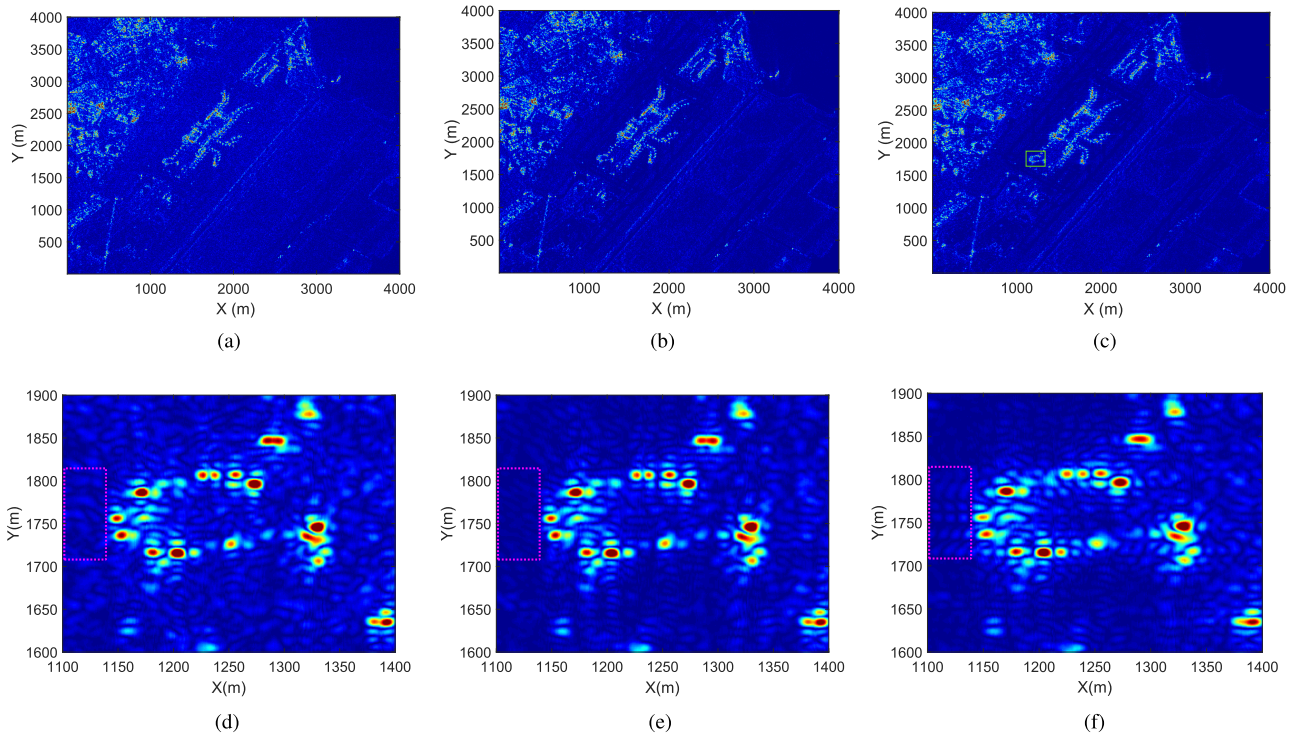


Fig. 15. Full target SAR images and corresponding enlarge image view of DFRC signals with $\gamma = 1, 2$ and LFM signal. (a) DFRC signal with $\gamma = 1$. (b) DFRC signal with $\gamma = 2$. (c) LFM. (d) Enlarge image view of Fig. 15(a). (e) Enlarge image view of Fig. 15(b). (f) Enlarge image view of Fig. 15(c).

gaps between curves are decreasing versus γ especially for a larger L, which means that the optimization limit is gradually reached. As a consequence, L should be carefully selected to balance the radar performance and communication rate.

To proceed further, let us observe the influence of SNR of (14) with different settings of u_1, u_2, and γ on the performance of communication, where the BER of 1000 Monte Carlo experiments with random communication information $\boldsymbol{w}$ versus SNR is shown in Fig. 11. Without loss of generality, let us set $L = 84$. Therein, curves are overlapped with $u_1 = u_2 = 1$ for different γ as $g_c(i) \in \{\pm 1, \pm j\}, i = 1, 2, \ldots, L$ is restricted, which also coincide with the curve of $u_1 = 0.5, u_2 = 1.5$ with $\gamma = 1$. Moreover, increasing of γ not only suppresses the PSL for radar performance but also provides the BER improvements when $u_1 = 0.5, u_2 = 1.5$.

C. Application in Joint SAR Imaging and Communication

In this section, let us apply the proposed DFRC scheme to joint SAR imaging through back projection (BP) algorithm[3] and communication with parameter settings listed in Table II. The gray icon of UESTC shown in Fig. 12 is transmitted for communication with $u_1 = 0.5$, $u_2 = 1.5$, $L = 82$ and 536 pulses required. Hence, the amount of information for Fig. 12 is 43 952 bits, and the data rate is 246 kb/s. In this respect, a DFRC system with a two-sided bandwidth of 256 MHz and a pulselength of 1 μs (leading to $N = 256$) is considered. Note that the channel is limited to a simplified, line-of-sight (LOS) model, and thus α_0 is governed by Rician fading. LFM signals with the same bandwidth and time duration but opposite chirp rates are adopted for channel estimation and start-stop synchronism. Without loss of generality, these 538 pulses are transmitted repeatedly.

[3]Other SAR imaging techniques can also be adopted [51], [52], [53]. However, it is out of the scope of this article.

TABLE II
SIMULATION PARAMETERS

System parameters	Value
Carrier frequency	9.6 GHz
PRF	3 kHz
Range sampling number	2048
Azimuth sampling number	2000
Transmitter initial position	[0 -33.33 3000] (m)
Transmitter velocity	[0 100 0] (m/s)

Fig. 13(a)–(f) depicts the recovered image received by the UE for SNR = 5,11,17 dB and $\gamma = 1, 2$, respectively. Since Fig. 13 is the intuitive visualization of Fig. 11, it clearly shows that better demodulation performance can be derived by increasing γ and SNR with $u_1 = 0.5$ and $u_2 = 1.5$. Besides, Fig. 13(f) is the same as the transmitted image with error-free demodulation realized.

Fig. 14(a)–(d) shows the point-target imaging simulation results using the designed waveforms with $\gamma = 1, 2$, LFM signal with the same bandwidth and time duration, and corresponding range profile along $Y = 0$ m, respectively. As expected, the sidelobe near the main lobe of the proposed DFRC waveform is lower than that of the LFM signal.

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TABLE III
IMAGE FEATURE INDEXES

	Image contrast	Entropy
LFM	0.0110	14.3173
DFRC, $\gamma = 1$	0.0123	14.6401
DFRC, $\gamma = 2$	0.0116	14.2309

Furthermore, the Changi Airport in Singapore is imaged with the characteristics of this scattering field derived from a SAR image, where the gray scale intensity of each pixel is used as a surrogate for the reflectivity. Fig. 15(a)–(f) depicts the SAR image results of different signals and their enlarged image view of the same district circled by the green box in Fig. 15(c). Looking over the images unveils that increasing γ improves the sidelobe level and further the imaging performance. The superior performance with respect to LFM is apparent in the area circled by the magenta box in Fig. 15(d)–(f). Finally, some image feature indexes of Fig. 15(a)–(c) derived by different sets of waveforms are concluded in Table III, where higher image contrast can be derived by the synthesized DFRC signals.

VI. CONCLUSION

In this article, we have proposed a general watermarking framework that accommodates different radio functions with nonblind, semiblind, or blind demodulation processes. Specifically, the DFRC system that includes multisegments like integrated couse waveform design and radar sensing at the integrated platform, up/downconversion for carrier synchronization and communication information demodulation has been addressed as a typical example, where watermarking technology is of great referential value due to similar advantages in robustness, perceptual transparency, steganography, and reversible self-recovery desired by DFRC systems. To this end, a novel DFRC signal design method has been proposed with information embedded as the watermark, where the PSL of ACF is suppressed with energy, PAR constraints to ensure capability with hardware, and DFT value constraints for communication. Hence, resorting to the PMM framework, iterative procedures have been conceived to synthesize optimized DFRC waveforms, where closed-form solutions can be obtained in each step. Remarkably, the overall computational complexity of the proposed algorithm is polynomial with respect to both the code length and the amount of information per pulse. At the analysis stage, the performance of the synthesized DFRC waveform has been assessed in terms of numerical and experimental simulations. The results have clearly shown that the proposed framework is capable of achieving dual function simultaneously with trade-off analysis between radar and communication performance. Furthermore, the proposed DFRC waveform has been applied in SAR imaging with image transmission at the same time to further illustrate its effectiveness.

Future research tracks might consider the signal design and processing methods of watermark-based DFRC framework for mutual assistance between radar and communication function. Specifically, the information viewed as a watermark can be used for recognition of true echo and antijamming processing, and the radar processing can be used for channel state information estimation and then promotes the robust design of DFRC waveform [33].

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